Efficiently update density-based decompositions under single-element insertions and deletions

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Website #37 | Solver verdict: SOLVED | Verifier verdict: ISSUES | Shown: solver output

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Problem

Let $M = (V, \mathcal{I})$ be a fixed matroid, and let $V' \subseteq V$ vary under single-element insertions and deletions. For a nonempty $U \subseteq V$ with $\mathrm{rank}_M(U) > 0$, define

$$\rho_M(U) = \frac{|U|}{\mathrm{rank}_M(U)},$$

with the conventions $\rho_M(\emptyset) = 0$ and $\rho_M(U) = +\infty$ when $U \neq \emptyset$ and $\mathrm{rank}_M(U) = 0$.

For the restriction $M|_{V'}$, let

$$V' = U_1 \sqcup U_2 \sqcup \cdots \sqcup U_k$$

be its density-based decomposition: at each step one takes, in the current contracted matroid, the densest subset of maximum cardinality and contracts it. (Trailing U_j's may be empty; the nonempty part has length at most $k = \mathrm{rank}_M(V)$.)

The task is to maintain this decomposition exactly under a sequence of single-element insertions and deletions, using fewer expensive "find a densest set" computations than recomputing the whole decomposition from scratch after every update.

Alternative readings considered

The statement admits two natural readings.

- A very strong reading asks for a worst-case $o(k)$-update bound independent of how much of the decomposition actually changes.
- A weaker but still exact reading asks for an output-sensitive algorithm whose cost is proportional to the part of the decomposition that really changes, rather than to the full rank k.

I adopt the exact, general-matroid, oracle-based reading: an update algorithm is acceptable if it maintains the decomposition exactly and asymptotically reduces the number of densest-set computations from k to $h + \log k$ (insertion) or h (deletion), where h is the length of the rebuilt suffix.

Related literature found

- **Huang–Sellier** (BACKGROUND). They define essentially the same greedy density-based decomposition, note that it is a reformulation of the principal sequence, and remark that densest sets can be computed via submodular minimization; the consulted versions use insertion/deletion lemmas inside a local-search proof for DCSes, not as an exact dynamic maintenance data structure for the full decomposition.

- **Fujishige** (BACKGROUND). His survey on principal partitions was useful for orienting the problem and confirming that the decomposition is the principal-sequence viewpoint in different notation.
- **Iwata–Fleischer–Fujishige** (TOOL). Strongly polynomial submodular-function minimization gives a black-box implementation of a densest-subset primitive on any contracted minor.

Proof sketch

Write the current nonempty decomposition as

$$V' = U_1 \sqcup \cdots \sqcup U_t, \qquad S_i := U_1 \cup \cdots \cup U_i,$$

with densities

$$\lambda_i := \rho_{(M/S_{i-1})|_{V' \setminus S_{i-1}}}(U_i).$$

Because of the maximum-cardinality tie-breaking, the nonempty densities are strictly decreasing: $\lambda_1 > \lambda_2 > \cdots > \lambda_t$.

The key structural lemma is a *splice lemma.* Suppose I keep an old prefix $U_1, \ldots, U_{s-1}$, contract S_{s-1}, and recompute the decomposition of the new residual ground there, obtaining first new density β. If $\beta < \lambda_{s-1}$, then the old prefix is still valid, and the full new decomposition is exactly

$$U_1, \ldots, U_{s-1}, (\text{new suffix}).$$

The proof is a weighted-average argument plus the monotonicity of density under further contraction.

For a deletion, this immediately solves the problem: if $x \in U_p$, then only the suffix starting at U_p can change. Indeed every set in the post-deletion residual minor already existed before deletion, so its density is at most $\lambda_p < \lambda_{p-1}$; therefore the splice lemma applies with $s = p$.

For an insertion, the changed suffix may start earlier than the block where the new element is first spanned; a small counterexample with two parallel pairs shows this. The right invariant is therefore not "span position" but the *largest stable prefix.* For each s, test whether the old prefix $U_1, \ldots, U_{s-1}$ survives by computing only the *first* density β_s of the residual minor after contracting S_{s-1} and inserting the new element. The splice lemma gives the forward implication; the converse comes from observing that if the old prefix really survives, then the next new block B_1 must satisfy $\rho(B_1) < \lambda_{s-1}$, for otherwise $U_{s-1} \cup B_1$ would beat (or tie and strictly enlarge) U_{s-1}. Hence

$$\text{prefix } U_1, \ldots, U_{s-1} \text{ survives } \iff \beta_s < \lambda_{s-1}.$$

This predicate is monotone in s, so binary search finds the first changed block using only $O(\log k)$ densest-set computations. Rebuilding the suffix once then finishes the update.

Main result

Theorem 1. *Let $M = (V, \mathcal{I})$ be a fixed matroid. Maintain the nonempty part*

$$V' = U_1 \sqcup \cdots \sqcup U_t$$

of the density-based decomposition of $M|_{V'}$, together with the cumulative sets $S_i := U_1 \cup \cdots \cup U_i$ and block densities

$$\lambda_i = \rho_{(M/S_{i-1})|_{V' \setminus S_{i-1}}}(U_i), \qquad i = 1, \ldots, t.$$

Let T_{dense} denote the cost of computing, in a contracted minor $(M/A)|_R$, the unique maximum-cardinality densest subset of that minor.

Then there is an exact dynamic update algorithm with the following worst-case oracle complexity per update.

- ***Insertion*** $u \notin V'$***:*** *$O(\log k + h)$ densest-set computations.*
- ***Deletion*** $u \in V'$***:*** *$O(h)$ densest-set computations.*

Here h is the number of nonempty blocks in the rebuilt suffix of the new *decomposition, and $k = \operatorname{rank}_M(V)$. Thus the update cost is output-sensitive: only the changed suffix is recomputed, instead of recomputing all k stages from scratch.*

Proof

We first reduce to the loopless case.

If $L := \{e \in V : \operatorname{rank}_M(\{e\}) = 0\}$ is the set of loops of M, then for every active set V' the first block of the decomposition is exactly $V' \cap L$, since every nonempty subset of $V' \cap L$ has density $+\infty$, and the maximum-cardinality tie-break picks all active loops. After removing that block, the remaining problem is the loopless matroid $M \setminus L$ on the active nonloops $V' \setminus L$. Therefore it suffices to prove the theorem for loopless matroids; from now on I assume M loopless.

Lemma 1. *Let $A \subseteq B \subseteq V$, and let $X \subseteq V \setminus B$. Then*

$$\operatorname{rank}_{M/B}(X) \leq \operatorname{rank}_{M/A}(X),$$

and hence

$$\rho_{M/A}(X) \leq \rho_{M/B}(X)$$

whenever $X \neq \emptyset$.

Proof. By the rank formula for contraction,

$$\operatorname{rank}_{M/B}(X) = \operatorname{rank}_M(X \cup B) - \operatorname{rank}_M(B),$$

$$\operatorname{rank}_{M/A}(X) = \operatorname{rank}_M(X \cup A) - \operatorname{rank}_M(A).$$

Since $A \subseteq B \subseteq X \cup B$, submodularity of the rank function gives

$$\operatorname{rank}_M(X \cup B) - \operatorname{rank}_M(B) \leq \operatorname{rank}_M(X \cup A) - \operatorname{rank}_M(A).$$

This is the rank inequality. The density inequality follows because the numerator $|X|$ is unchanged and the denominator weakly decreases when more is contracted. $\square$

Lemma 2. *Let*

$$V' = U_1 \sqcup \cdots \sqcup U_t$$

be the nonempty part of the density-based decomposition of $M|_{V'}$, and set $S_i := U_1 \cup \cdots \cup U_i$. Then

$$\lambda_1 > \lambda_2 > \cdots > \lambda_t,$$

where $\lambda_i := \rho_{(M/S_{i-1})|_{V' \setminus S_{i-1}}}(U_i)$.

Proof. Fix $i < t$. In the matroid $(M/S_{i-1})|_{V'\setminus S_{i-1}}$, the set U_i is chosen as a densest subset of maximum cardinality. Suppose for contradiction that $\lambda_{i+1} \geq \lambda_i$.

Because U_{i+1} is measured after contracting U_i, we have

$$\mathrm{rank}_{M/S_{i-1}}(U_i \cup U_{i+1}) = \mathrm{rank}_{M/S_{i-1}}(U_i) + \mathrm{rank}_{M/S_i}(U_{i+1}).$$

Hence

$$\rho_{M/S_{i-1}}(U_i \cup U_{i+1}) = \frac{|U_i| + |U_{i+1}|}{\mathrm{rank}_{M/S_{i-1}}(U_i) + \mathrm{rank}_{M/S_i}(U_{i+1})},$$

which is the weighted average of λ_i and λ_{i+1}. Therefore:

- if $\lambda_{i+1} > \lambda_i$, then $\rho_{M/S_{i-1}}(U_i \cup U_{i+1}) > \lambda_i$, contradicting the choice of U_i;
- if $\lambda_{i+1} = \lambda_i$, then $\rho_{M/S_{i-1}}(U_i \cup U_{i+1}) = \lambda_i$, but $U_i \cup U_{i+1}$ is strictly larger than U_i, contradicting the maximum-cardinality tie-break.

So $\lambda_{i+1} < \lambda_i$. □

The next lemma is the core of the update algorithm.

Lemma 3 (splice lemma)**.** *Fix $s \in \{1, \dots, t+1\}$, and let $R \subseteq V \setminus S_{s-1}$. Let*

$$R = B_1 \sqcup \cdots \sqcup B_q$$

be the nonempty part of the density-based decomposition of $(M/S_{s-1})|_R$, and let

$$\beta := \rho_{(M/S_{s-1})|_R}(B_1)$$

if $q \geq 1$, while $\beta := 0$ if $q = 0$.

If either $s = 1$ or $\beta < \lambda_{s-1}$, then the density-based decomposition of $(M|_{S_{s-1} \cup R})$ is exactly

$$U_1, \dots, U_{s-1}, B_1, \dots, B_q.$$

Proof. I prove first that U_{s-1} is the correct first block after contracting S_{s-2} (when $s > 1$); then I move backward inductively through the prefix.

Assume $s > 1$. Put

$$N := (M/S_{s-2})|_{U_{s-1} \cup R}.$$

Let $Y \subseteq U_{s-1} \cup R$, and write

$$A := Y \cap U_{s-1}, \qquad C := Y \cap R.$$

Then

$$|Y| = |A| + |C|, \qquad \mathrm{rank}_N(Y) = \mathrm{rank}_N(A) + \mathrm{rank}_{N/A}(C).$$

Because U_{s-1} was the maximum-cardinality densest subset in the old residual minor $(M/S_{s-2})|_{V'\setminus S_{s-2}}$, every subset of U_{s-1} has density at most λ_{s-1}. Hence

$$|A| \leq \lambda_{s-1}\,\mathrm{rank}_N(A).$$

Next, by Lemma 1,

$$\rho_{N/A}(C) \leq \rho_{N/U_{s-1}}(C).$$

But $N/U_{s-1} = (M/S_{s-1})|_R$, whose densest density is β. Therefore

$$|C| \le \beta \operatorname{rank}_{N/A}(C).$$

If $C \neq \emptyset$, then $\beta < \lambda_{s-1}$, so in fact

$$|C| < \lambda_{s-1} \operatorname{rank}_{N/A}(C).$$

Adding the inequalities,

$$|Y| < \lambda_{s-1}\big(\operatorname{rank}_N(A) + \operatorname{rank}_{N/A}(C)\big) = \lambda_{s-1} \operatorname{rank}_N(Y)$$

whenever $C \neq \emptyset$. So every densest subset of N is contained in U_{s-1}.

Among subsets of U_{s-1}, the unique maximum-cardinality densest subset is U_{s-1} itself, because the old decomposition chose U_{s-1} with exactly that tie-breaking rule. Hence the first block of the new decomposition in N is still U_{s-1}.

After contracting U_{s-1}, the remaining active ground is exactly R, and by assumption its decomposition is $B_1, \dots, B_q$.

It remains to justify the earlier blocks $U_{s-2}, \dots, U_1$. This is now immediate by backward induction. Indeed, after fixing $U_{i+1}, \dots, U_{s-1}$, the first density in the remainder after contracting S_i is

$$\lambda_{i+1} < \lambda_i \quad (i < s-1)$$

by Lemma 2. Repeating the same argument as above shows that U_i remains the maximum-cardinality densest subset after contracting S_{i-1}. Therefore the whole prefix survives, and the full new decomposition is the stated concatenation.

The case $s = 1$ is tautological: there is no prefix to preserve. □

The converse criterion needed for insertion is equally elementary.

Lemma 4 (stability criterion for insertion)**.** *Let $W \subseteq V$ be the current active set with decomposition $U_1, \dots, U_t$, and let $u \notin W$. For $s \in \{1, \dots, t+1\}$, define*

$$R_s^+ := (W \setminus S_{s-1}) \cup \{u\},$$

and let β_s^+ be the density of the first block in the decomposition of $(M/S_{s-1})|_{R_s^+}$.

Then the following are equivalent.

1. *The first $s-1$ old blocks $U_1, \dots, U_{s-1}$ remain the first $s-1$ blocks after inserting u.*

2. *Either $s = 1$, or $\beta_s^+ < \lambda_{s-1}$.*

Proof. (2) $\Rightarrow$ (1) is exactly Lemma 3 with $R = R_s^+$.

For (1) $\Rightarrow$ (2), assume $s > 1$, and let B_1 be the first new block after the surviving prefix $U_1, \dots, U_{s-1}$. Then, by definition,

$$\beta_s^+ = \rho_{(M/S_{s-1})|_{R_s^+}}(B_1).$$

In the matroid $(M/S_{s-2})|_{U_{s-1} \cup R_s^+}$, the rank of $U_{s-1} \cup B_1$ is

$$\operatorname{rank}_{M/S_{s-2}}(U_{s-1}) + \operatorname{rank}_{M/S_{s-1}}(B_1),$$

so

$$\rho_{M/S_{s-2}}(U_{s-1} \cup B_1) = \frac{|U_{s-1}| + |B_1|}{\mathrm{rank}_{M/S_{s-2}}(U_{s-1}) + \mathrm{rank}_{M/S_{s-1}}(B_1)},$$

the weighted average of λ_{s-1} and β_s^+.

If $\beta_s^+ > \lambda_{s-1}$, this weighted average is $> \lambda_{s-1}$, contradicting that U_{s-1} is the densest set at that stage. If $\beta_s^+ = \lambda_{s-1}$, then $U_{s-1} \cup B_1$ has the same density as U_{s-1} but larger cardinality, contradicting the maximum-cardinality tie-break. Hence $\beta_s^+ < \lambda_{s-1}$. □

Lemma 5 (deletion starts at the deleted block)**.** *Let $x \in U_p$. Define*

$$R_p^- := (W \setminus S_{p-1}) \setminus \{x\},$$

and let β_p^- be the first density in the decomposition of $(M/S_{p-1})|_{R_p^-}$. Then

$$\beta_p^- \le \lambda_p.$$

In particular, if $p > 1$, then

$$\beta_p^- < \lambda_{p-1}.$$

Hence the new decomposition after deleting x is

$$U_1, \ldots, U_{p-1}, \textit{(decomposition of } (M/S_{p-1})|_{R_p^-}\textit{)}.$$

Proof. Every subset of R_p^- is also a subset of the old residual ground $W \setminus S_{p-1}$. Its rank in the contracted matroid (M/S_{p-1}) is unchanged by the deletion of x, since $x \notin R_p^-$. Therefore every candidate set in the post-deletion residual minor had exactly the same density before deletion. Since U_p was the densest set in the old residual minor, every such density is at most λ_p. Thus $\beta_p^- \le \lambda_p$.

By Lemma 2, if $p > 1$ then $\lambda_p < \lambda_{p-1}$. So $\beta_p^- < \lambda_{p-1}$, and Lemma 3 applies with $s = p$. □

Now I state the algorithm.

Data maintained.

- the current nonempty blocks $U_1, \ldots, U_t$;
- the cumulative sets S_i;
- the densities λ_i;
- a map $\mathrm{pos} : W \to \{1, \ldots, t\}$ sending each active element to its current block.

Primitive. For a contracted minor $(M/A)|_R$, let $\mathrm{Dense}(A, R)$ return its maximum-cardinality densest subset D and its density $\rho_{(M/A)|_R}(D)$.

Insertion of $u \notin W$. For each $s \in \{1, \ldots, t+1\}$, let

$$\mathrm{stable}(s) := \begin{cases} \text{true}, & s = 1, \\ \left(\beta_s^+ < \lambda_{s-1}\right), & s > 1, \end{cases}$$

where β_s^+ is obtained by one call to $\mathrm{Dense}(S_{s-1}, R_s^+)$ with $R_s^+ = (W \setminus S_{s-1}) \cup \{u\}$.

By Lemma 4, $\mathrm{stable}(s)$ holds if and only if the first $s-1$ old blocks survive. Therefore $\mathrm{stable}(s)$ is monotone in s, and binary search finds

$$s^\star = \max\{s : \mathrm{stable}(s)\}.$$

Then rebuild exactly once the decomposition of $(M/S_{s^\star-1})|_{R^+_{s^\star}}$ by repeated calls to Dense, say

$$R^+_{s^\star} = B_1 \sqcup \cdots \sqcup B_h.$$

Replace the old suffix $U_{s^\star}, \ldots, U_t$ by $B_1, \ldots, B_h$. By Lemma 3, this is the correct new decomposition.

Deletion of $x \in W$. Let $p := \mathrm{pos}(x)$. Rebuild the decomposition of

$$(M/S_{p-1})|_{R^-_p}, \qquad R^-_p = (W \setminus S_{p-1}) \setminus \{x\},$$

again by repeated calls to Dense, say

$$R^-_p = B_1 \sqcup \cdots \sqcup B_h.$$

Replace the old suffix $U_p, \ldots, U_t$ by $B_1, \ldots, B_h$. By Lemma 5, this is exactly the new decomposition.

Correctness. Insertion is correct by Lemma 4 (binary search finds the largest surviving prefix) followed by Lemma 3. Deletion is correct by Lemma 5.

Complexity. Let h be the number of nonempty blocks in the rebuilt suffix.

- Insertion uses $O(\log(t+1)) = O(\log k)$ calls to Dense for the binary search, and then h more calls to rebuild the suffix.
- Deletion uses h calls to Dense.

Bookkeeping (updating the arrays of blocks, cumulative sets, and pos) is linear in the number of elements whose block assignment changes, which is unavoidable if the explicit partition is to be stored.

This proves the theorem. □

Gap to the full statement

Under the exact oracle-based reading fixed above, there is no gap: the theorem gives an exact dynamic maintenance algorithm. The bound is output-sensitive rather than worst-case $o(k)$; in the worst case one may still need to rebuild the whole suffix.

Self red-team

Counterexample attempts.

1. *Backward propagation on insertion.* Take a direct sum of two rank-1 parallel classes:

$$U_1 = \{a, b\}, \qquad U_2 = \{c\},$$

and insert u parallel to c. Then the true new first block is $\{a, b, c, u\}$. This breaks the naive idea "start from the block where u is first spanned and never move left." Our algorithm survives this test because for $s = 2$ the residual first density equals λ_1, so stable(2) is false by the equality case.

2. *Splitting an old block on insertion.* Let the old active set be a single uniformly dense rank-3 block on 4 elements (a $U_{3,4}$-minor), and insert a new element parallel to one old element. The old first block must be rebuilt from scratch. Our algorithm returns $s^\star = 1$, because every $s > 1$ fails the stability test.

3. *Backward propagation on deletion.* I tried to force a deletion in a later block to disturb an earlier block by making the later block denser after deletion. This cannot happen: every post-deletion candidate subset already existed pre-deletion with the same density, so the residual densest density never increases. This is exactly Lemma 5.

Three likely referee objections.

1. *"What about sets that use only part of U_{s-1} together with outside elements?"* This is the delicate point in Lemma 3. It is handled by Lemma 1: contracting only a subset $A \subseteq U_{s-1}$ can only make outside sets *less* dense than contracting all of U_{s-1}, where the density is already $< \lambda_{s-1}$.

2. *"Why is equality $\beta = \lambda_{s-1}$ forbidden?"* Because with the full U_{s-1} contracted, rank additivity is exact:
$$r(U_{s-1} \cup B_1) = r(U_{s-1}) + r_{M/S_{s-1}}(B_1).$$
Hence $U_{s-1} \cup B_1$ has the same density as U_{s-1} and strictly larger cardinality, contradicting the maximum-cardinality tie-break.

3. *"How are loops handled?"* They are separated off at the start. The active loops always form the unique first block of density $+\infty$, and the remaining problem is loopless.

Boundary checks.

- $V' = \emptyset$: insertion gives $s^\star = 1$ and rebuilds the singleton suffix.
- $s^\star = t+1$: the entire old decomposition survives and the inserted element appends at the end (or merges with the trailing density-1 block if equality forces that).
- Deletion may produce an empty rebuilt suffix; then the remaining blocks are just the preserved prefix.
- I verified that the proof does *not* claim worst-case sublinear update time in k; worst-case $h = k$ is still possible.

Invoked results

- **Strongly polynomial submodular-function minimization** (TOOL). *Statement.* Every submodular set function $f : 2^E \to \mathbb{Z}$ on a finite ground set can be minimized in strongly polynomial time. *Citation.* Iwata, Fleischer, and Fujishige (JACM 2001), DOI 10.1145/502090.502096.

  Hypothesis audit. For a fixed contraction set $A \subseteq V$, active residual set $R \subseteq V \setminus A$, and rational parameter $\tau = p/q > 0$, define
$$f_{A,R,\tau}(X) := q\big(\mathrm{rank}_M(A \cup X) - \mathrm{rank}_M(A)\big) - p|X|, \qquad X \subseteq R.$$

The map $X \mapsto \mathrm{rank}_M(A \cup X) - \mathrm{rank}_M(A)$ is a matroid rank function on R, hence submodular; $X \mapsto |X|$ is modular; therefore $f_{A,R,\tau}$ is submodular. The ground set R is finite. So the theorem applies.

A call to Dense(A, R) can therefore be implemented exactly by parametric search on τ, minimizing $f_{A,R,\tau}$ for candidate densities. This implementation detail is not used inside the correctness proof; it only justifies that the densest-set primitive is polynomial-time computable in the general matroid setting.

Self-confidence

3 — I checked the splice criterion, the insertion converse, the deletion monotonicity, and the equality/tie-breaking cases carefully; the result is exact under the stated oracle model, though the output-sensitive interpretation of "more efficiently" is the one I solved.

Bibliography


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\begin{verbatim}
@article{IwataFleischerFujishige2001,
  author    = {Satoru Iwata and Lisa Fleischer and Satoru Fujishige},
  title     = {A Combinatorial, Strongly Polynomial-Time Algorithm for Minimizing
               Submodular Functions},
  journal   = {Journal of the ACM},
  volume    = {48},
  number    = {4},
  pages     = {761--777},
  year      = {2001},
  doi       = {10.1145/502090.502096}
}

@inproceedings{HuangSellier2024,
  author    = {Chien-Chung Huang and Francois Sellier},
  title     = {Robust Sparsification for Matroid Intersection with Applications},
  booktitle = {Proceedings of the 2024 Annual ACM-SIAM Symposium on Discrete
               Algorithms (SODA)},
  pages     = {2916--2940},
  year      = {2024},
  doi       = {10.1137/1.9781611977912.104},
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  archivePrefix = {arXiv}
}

@incollection{Fujishige2009,
  author    = {Satoru Fujishige},
  title     = {Theory of Principal Partitions Revisited},
  booktitle = {Research Trends in Combinatorial Optimization},
  editor    = {William J. Cook and Laszlo Lovasz and Jens Vygen},
  pages     = {127--162},
  publisher = {Springer},
  year      = {2009},
  doi       = {10.1007/978-3-540-76796-1_7}
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